

Capacitance of a PN Junction

Objective

Measure the capacitance of a PN junction under various dc bias voltage, and find the zero-bias junction capacitance C_{j0} .

Theory - AC Analysis

In this laboratory we will design an experiment to measure the capacitance of a PN junction as a function of the dc bias voltage V across the device. The constraints (reverse-biased) put upon this design are that the dc voltage should be varied from -20 V to +0.0 V. Some factors should be kept in mind:

- 1) Typical junction capacitance are very small (about 100 pF), so we must make sure that we can measure these small capacitance
- 2) High frequencies must be used (you will have to justify this in your report)
- 3) Circuit leads (wires) may add stray capacitance to our measurements

An example of such a design is now presented, but a different design for achieving this task is also welcomed. The circuit chosen appears in Figure 1. **The diode is replaced with a parallel combination of a resistor and a capacitor for analysis (but not for measurements).** The ac and dc equivalent circuits are shown in Figure 2.

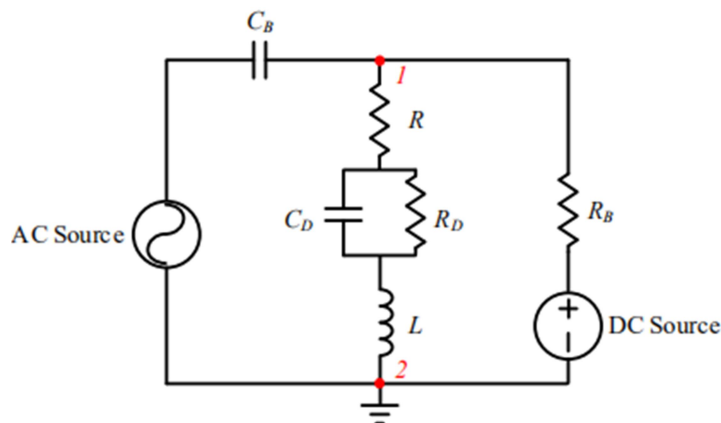


Figure 1. Experimental set-up with the diode replaced with its RC diode model

Ignoring the small R_D , If we calculate the total impedance of the RLC combination from point 1 to point 2 in Figure 1, taking into account both the real and imaginary parts, we get,

$$Z_{eq} = R + \frac{1}{j\omega C_D} + j\omega L \quad (1)$$

Rearranging and combining the real and imaginary terms,

$$Z_{eq} = R + j\left(\omega L - \frac{1}{\omega C_D}\right) \quad (2)$$

From this we can clearly see that resonance will occur for,

$$C_D = \frac{1}{(2\pi f)^2 L} \quad (3)$$

This is derived from the fact that the magnitude of a complex number has a minimum value when its imaginary component is equal to zero. Since the value of the inductance L is fixed, the only variable is frequency which is a convenient controlling parameter. So that we can calculate C_D easily according to (3) after we find the resonant frequency.

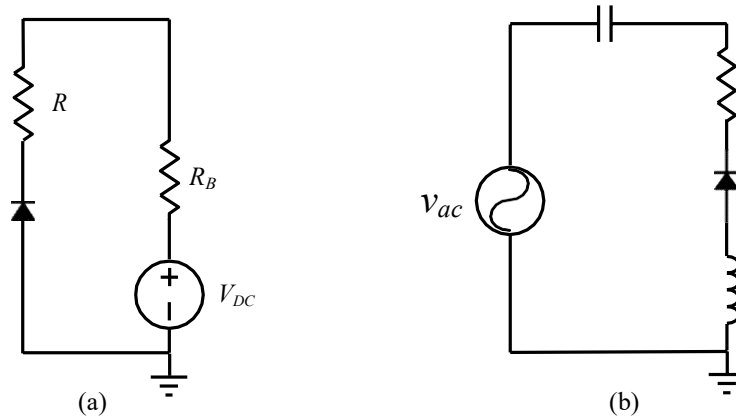


Figure 2. (a) DC equivalent circuit; (b) AC equivalent circuit.

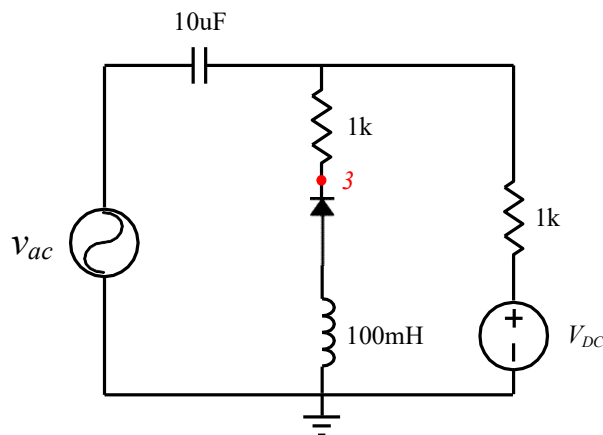


Figure 3. Circuit for capacitance measurement

Reference

- [1] Drexel University, ECE-E302, Electronic Devices.

Diode Capacitance Measurements

Reverse bias the diode with voltage from 1V to 5V with 1V step and measure the capacitance of the reverse-biased PN junction. Note: The capacitance measurement will be performed using the Bode Analyzer. Fit the capacitance with the step junction model and calculate the zero bias junction capacitance C_{j0} .